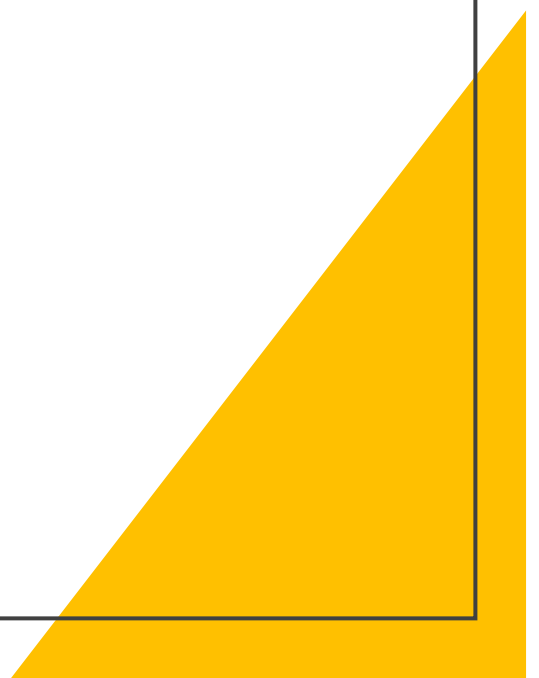


Electron Correlation Methods

Lecture 5

Liviu Ungur



Densities and distributions

Density refers to a continuous spatial distribution of a certain property.
Commonly encountered densities of interest are:

Probability density

$$p(x, y, z) = |\psi(x, y, z)|^2$$

Electron density

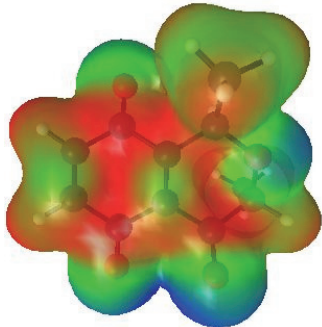
$$\rho(\vec{r}) = N \int d^3\vec{r}_2 \int d^3\vec{r}_3 \cdots \int d^3\vec{r}_N |\Psi(\vec{r}, \vec{r}_2, \dots, \vec{r}_N)|^2$$

Spin density

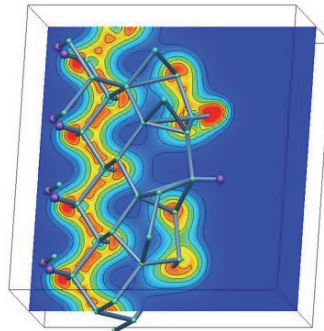
$$\sigma(\vec{r}) = \rho_{\uparrow}(\vec{r}) - \rho_{\downarrow}(\vec{r})$$

Charge density

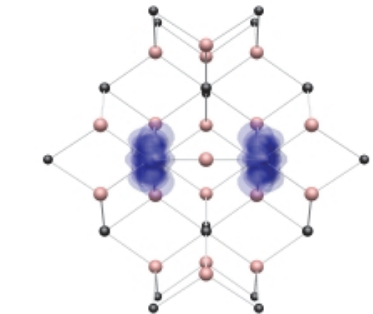
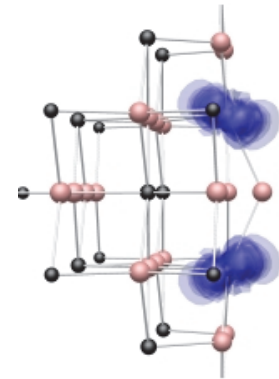
$$\rho_C(\vec{r}) = Ne \int d^3\vec{r}_2 \int d^3\vec{r}_3 \cdots \int d^3\vec{r}_N |\Psi(\vec{r}, \vec{r}_2, \dots, \vec{r}_N)|^2$$



Electron isodensity surface mapped with electrostatic potential.



Charge density map of a modified silicon surface.



Spin density around an oxygen vacancy in cerium oxide.

Lecture (≈ 1 h)

- Density Functional Theory
- Hohenberg-Kohn theorem
- Energy as a functional of electronic density
- Density functionals: LDA, GGA, hybrid (B3LYP), meta-GGA, etc.
- Practical aspects: numerical integration, basis set size, etc.

Electron density distribution

The **total electron density** can be obtained as a sum of probability distributions of each individual electron:

$$\rho(\vec{r}) = \sum_{i=1}^N |\chi_i(\vec{r})|^2 \quad \chi_i(\vec{r}) = \sum_{\mu=1}^K c_{\mu i} \phi_{\mu}(\vec{r})$$

Where N is the number of electrons in the system and K is the number of functions in the basis. Therefore, the **total density** can be expressed as:

$$\rho(\vec{r}) = \sum_{\mu, \nu=1}^K \sum_{i=1}^N c_{\mu i}^* c_{\nu i} \phi_{\mu}^*(\vec{r}) \phi_{\nu}(\vec{r})$$

Integrating this over all space gives the total number of electrons in the system:

$$N = \int \rho(\vec{r}) d^3\vec{r} = \sum_{\mu, \nu=1}^K \sum_{i=1}^N c_{\mu i}^* c_{\nu i} \int \phi_{\mu}^*(\vec{r}) \phi_{\nu}(\vec{r}) d^3\vec{r}$$

Two shorthands are commonly defined in this expression:

$$P_{\mu\nu} = \sum_{i=1}^N c_{\mu i}^* c_{\nu i} \quad S_{\mu\nu} = \int \phi_{\mu}^*(\vec{r}) \phi_{\nu}(\vec{r}) d^3\vec{r} \quad N = \sum_{\mu, \nu=1}^K P_{\mu\nu} S_{\mu\nu}$$

"density matrix" "overlap matrix"

Electron density distribution

We can now find out how much a particular pair of orbitals contributes to the total observable electron density:

$$N_{\mu\nu} = \sum_{i=1}^N \underbrace{c_{\mu i}^* c_{\nu i}}_{\text{orbital populations}} \underbrace{\int \varphi_{\mu}^*(\vec{r}) \varphi_{\nu}(\vec{r}) d^3\vec{r}}_{\text{orbital overlap}} = P_{\mu\nu} S_{\mu\nu}$$

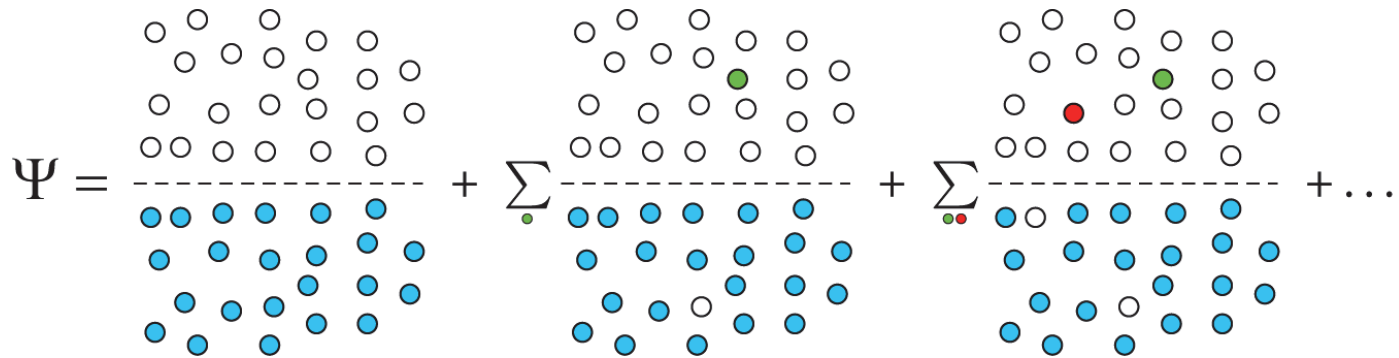
This density partitioning into pairs of orbitals is useful in charge and bonding structure analysis. In atom-centred basis sets there are two possible situations:

1. Both φ_{μ} and φ_{ν} are located on the same atom. All of the density in $N_{\mu\nu}$ can be assigned to this atom and interpreted as contributing to its charge.
2. The two orbitals φ_{μ} and φ_{ν} are located on different atoms. The density in $N_{\mu\nu}$ is then shared between the two atoms and can be interpreted as contributing to a chemical bond as well as charges.

This formalism can be generalized to partition spin as well as charge:

$$P_{\mu\nu}^{\alpha} = \sum_{i=1}^N c_{\mu i}^{\alpha*} c_{\nu i}^{\alpha} \quad P_{\mu\nu}^{\beta} = \sum_{i=1}^N c_{\mu i}^{\beta*} c_{\nu i}^{\beta}$$

Electron Correlation Energy



$$\Psi = a_0 \psi_{HF} + \sum_S a_S \psi_S + \sum_D a_D \psi_D + \sum_T a_T \psi_T + \dots$$

HF energy:

$$E_{HF} = \langle \psi_{HF} | \hat{H} | \psi_{HF} \rangle$$

Exact Energy:

$$E_{exact} = \langle \Psi | \hat{H} | \Psi \rangle$$

Correlation Energy: $E_{corr} = E_{exact} - E_{HF}$

Usually, E_{corr} is only about 1-5% of the E_{exact} but it is very important for applications

Density Functional Theory

Within the Born-Oppenheimer Hamiltonian, the type of molecule or solid is defined by the electron-nucleus interaction :

$$\begin{aligned}\hat{V}_{ext} &= -\frac{1}{4\pi\epsilon_0} \sum_{i=1}^{N_{elec}} \sum_{j=1}^{N_{nuc}} \frac{e^2 Z_j}{|\vec{R}_j - \vec{r}_i|} = && \text{SI units} \\ &= -\sum_{i=1}^{N_{elec}} \sum_{j=1}^{N_{nuc}} \frac{Z_j}{|\vec{R}_j - \vec{r}_i|} && \text{atomic units}\end{aligned}$$

(this is often called the 'external potential')

Density Functional Theory

Electron density operator for one single electron :

$$\hat{\rho}(\vec{r}) = \delta(\vec{r}' - \vec{r})$$

Electron density for one single electron:

$$\rho(\vec{r}) = \langle \Psi | \hat{\rho}(\vec{r}) | \Psi \rangle = \int \Psi^*(\vec{r}') \Psi(\vec{r}) \delta(\vec{r}' - \vec{r}) d\vec{r}' = \Psi^*(\vec{r}) \Psi(\vec{r}) = |\Psi(\vec{r})|^2$$

Interpretation: Probability to find the electron at position $\vec{r}$

Density Functional Theory

Electron density operator for N electrons:

$$\hat{\rho}(\vec{r}) = \sum_{i=1}^N \delta(\vec{r} - \vec{r}_i)$$

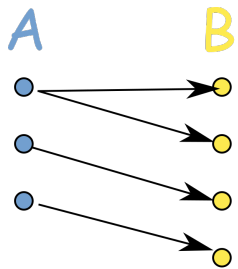
Electron density for N electron:

$$\begin{aligned} \rho(\vec{r}) &= \langle \Psi | \hat{\rho}(\vec{r}) | \Psi \rangle = \sum_{i=1}^N \int \Psi^*(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_i \equiv \vec{r}, \dots, \vec{r}_N) \Psi(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_i \equiv \vec{r}, \dots, \vec{r}_N) d\vec{r}_1 d\vec{r}_2 \dots d\vec{r}_i \dots d\vec{r}_N \\ &= \sum_{i=1}^N |\varphi(\vec{r}_i)|^2 \end{aligned}$$

Interpretation: Probability to find any electron at position $\vec{r}$, regardless where all other $N-1$ electrons are.

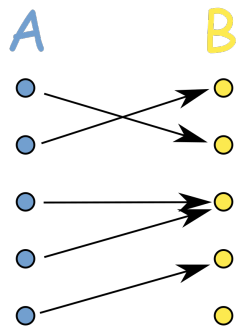
$\rho(\vec{r}) \equiv \rho(x, y, z)$ is a real function of 3 variables regardless of the size of N .

Density Functional Theory



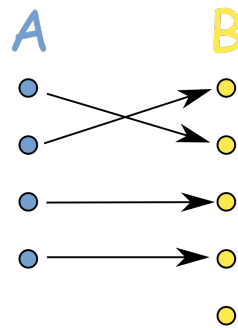
NOT a
Function

A has many B



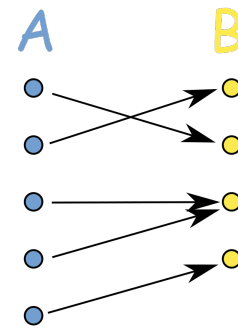
General
Function

B can have many A



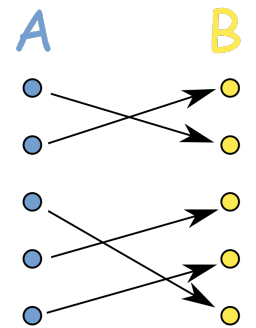
Injective
(not surjective)

B can't have many A



Surjective
(not injective)

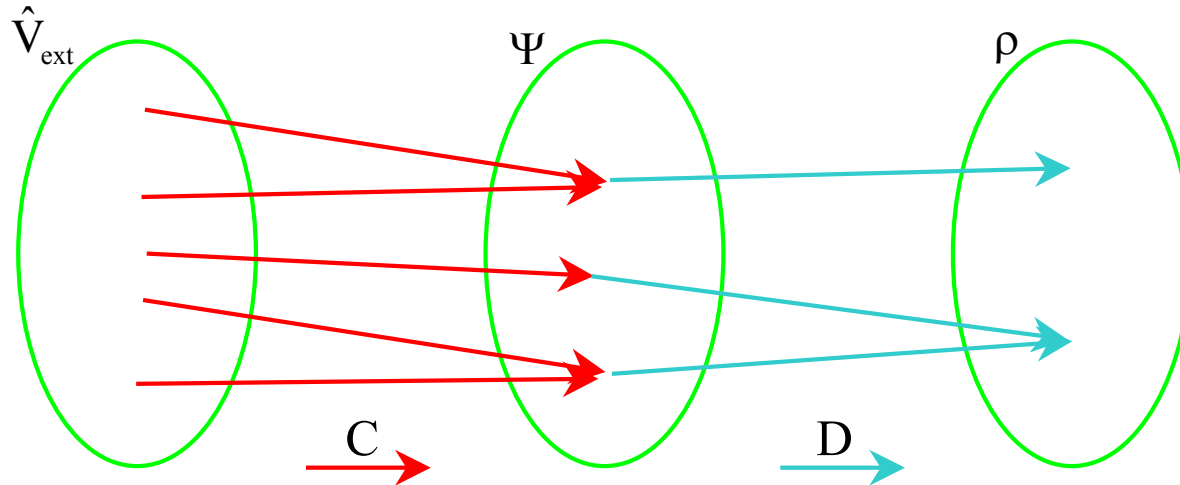
Every B has some A



Bijjective
(injective, surjective)

A to B, perfectly

First Theorem of Hohenberg and Kohn

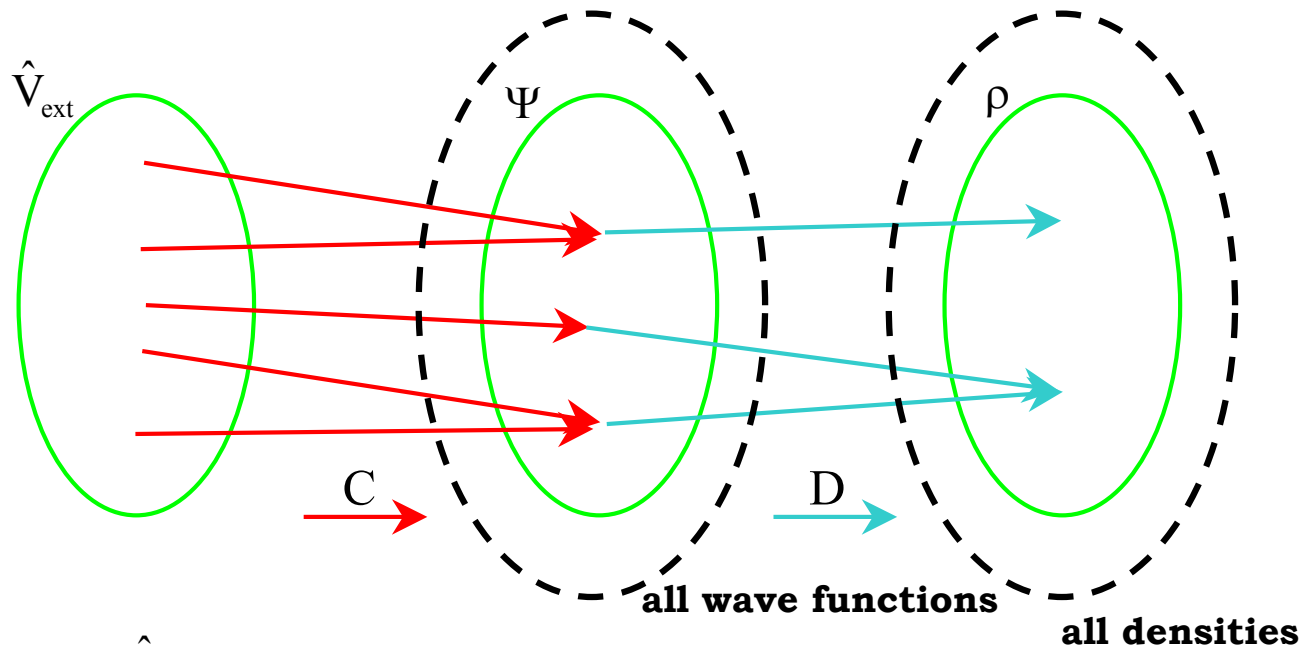


$C: \hat{V}_{\text{ext}} \mapsto \Psi$ surjective by construction

$D: \Psi \mapsto \rho$ surjective by construction

Are C and D also injective ?

First Theorem of Hohenberg and Kohn



$$C: \hat{V}_{\text{ext}} \mapsto \Psi$$

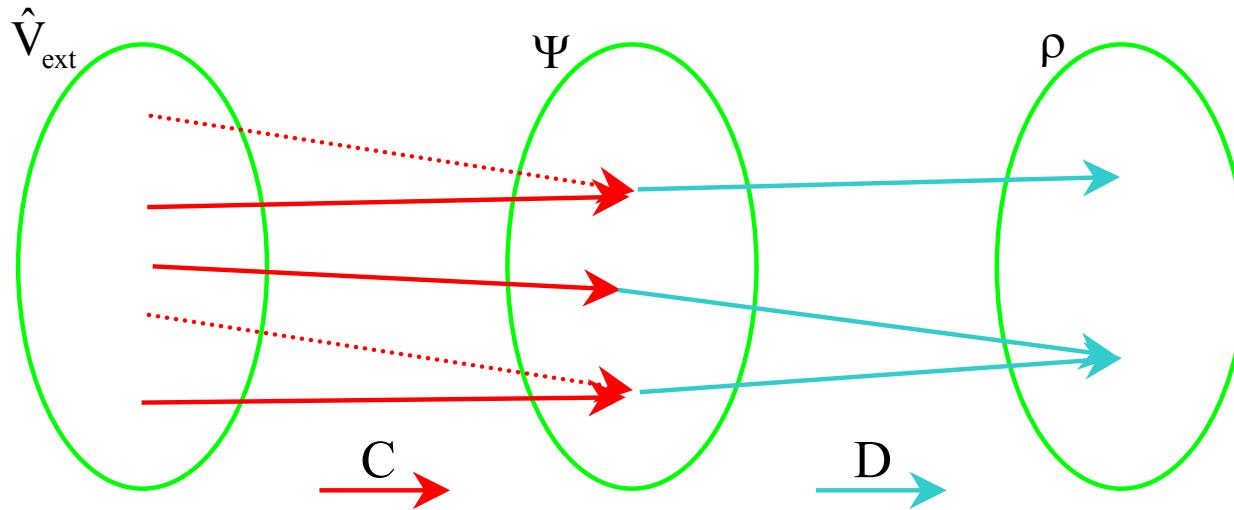
surjective by construction

$$D: \Psi \mapsto \rho$$

surjective by construction

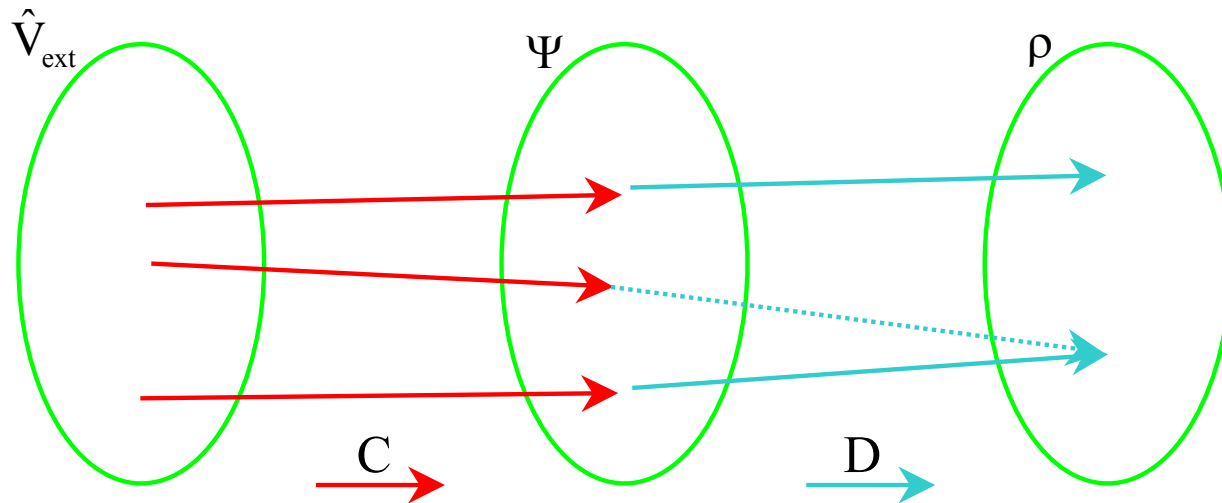
Are C and D also injective ?

First Theorem of Hohenberg and Kohn



$C: \hat{V}_{\text{ext}} \mapsto \Psi$ injective (no surprise)

First Theorem of Hohenberg and Kohn



$C: \hat{V}_{\text{ext}} \mapsto \Psi$ injective (no surprise)

$D: \Psi \mapsto \rho$ injective (very surprising !)
= first theorem of Hohenberg and Kohn (1964)

Proof of the First Theorem of Hohenberg and Kohn

We wish to prove: if $\Psi \neq \Psi'$, then $\rho \neq \rho'$

We do this by assuming that $\Psi \neq \Psi'$, then $\rho = \rho'$, which will lead to contradiction

Ψ is the ground state of a Hamiltonian H , ground state energy is E_{gs}

Ψ' is the ground state of a Hamiltonian H' , ground state energy is E'_{gs}

V and V' are external potentials for the two cases:

Same number of electrons (otherwise trivial), hence:

$$H = T + V \quad \text{and} \quad H' = T + V' \Rightarrow H = H' - V' + V$$

Due to variational principle:

$$E_{gs} = \langle \Psi | H | \Psi \rangle < \langle \Psi' | H | \Psi' \rangle = \langle \Psi' | H' - V' + V | \Psi' \rangle = \langle \Psi' | H' | \Psi' \rangle + \langle \Psi' | V - V' | \Psi' \rangle \Rightarrow$$

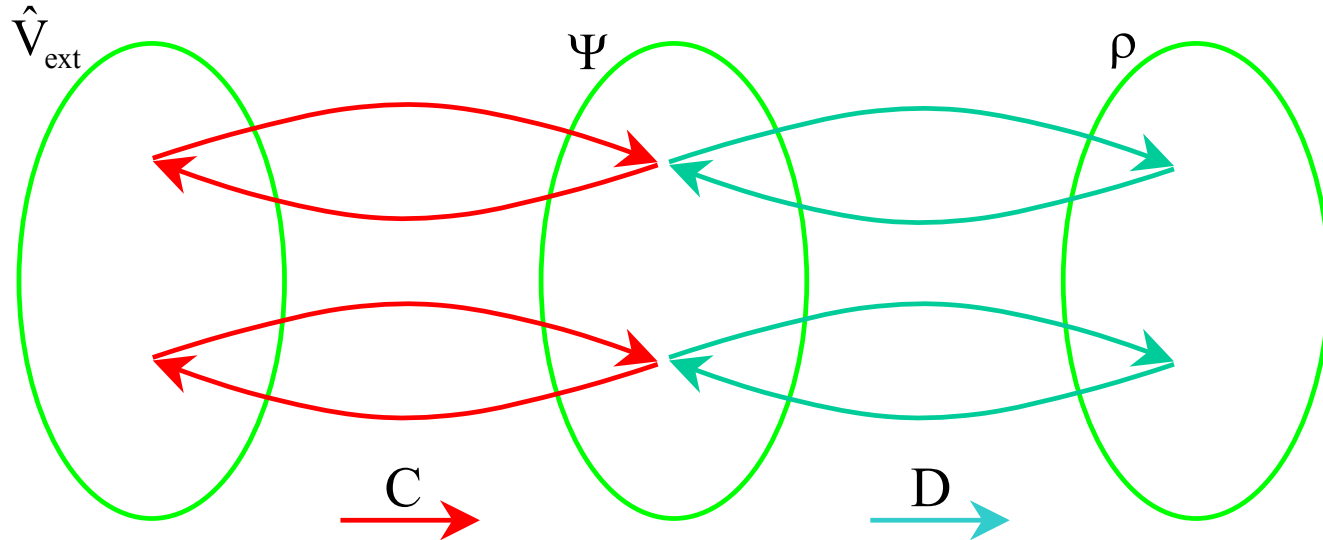
$$E_{gs} < E'_{gs} + \int \rho'(\vec{r}) [V(\vec{r}) - V'(\vec{r})] d(\vec{r})$$

Repeat argument, starting with Ψ' :

$$E'_{gs} < E_{gs} - \int \rho(\vec{r}) [V(\vec{r}) - V'(\vec{r})] d(\vec{r})$$

Add up the two expressions, assuming that $\rho = \rho' \Rightarrow E_{gs} + E'_{gs} < E'_{gs} + E_{gs}$ contradiction.

First Theorem of Hohenberg and Kohn



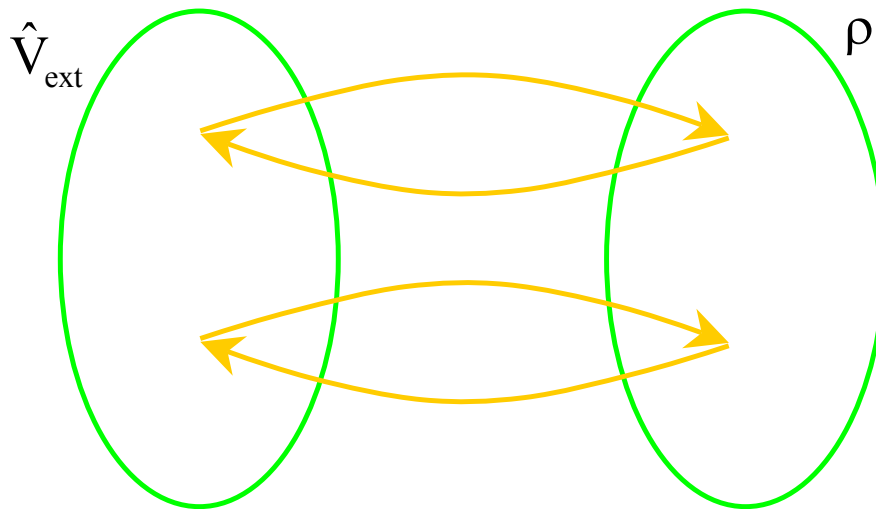
$$C: \hat{V}_{\text{ext}} \mapsto \Psi \quad \text{bijective}$$

$$D: \Psi \mapsto \rho \quad \text{bijective}$$

$\Rightarrow$ one-to-one relations

First Theorem of Hohenberg and Kohn

*“ There is a **one-to-one** correspondence between the ground state density of a many-electron system and the external potential. ”*



Consequence: any observable ground state property of the many-electron system can be written as a **unique** functional of the density :

$$\text{property} = \langle \Psi | \hat{O} | \Psi \rangle = O[\rho]$$

What is a *functional*?

A function maps *numbers* onto (complex) numbers:

$$f: \mathbb{R} \rightarrow \mathbb{C}: x \mapsto f(x)$$

A functional maps *functions* onto (complex) numbers :

$$f: \mathfrak{F} \rightarrow \mathbb{C}: x \mapsto F[f]$$

Example:

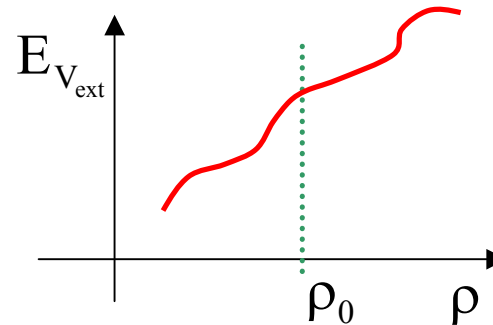
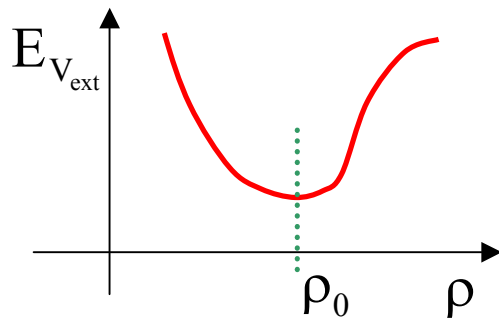
$$F[f] = \int_{-\infty}^{+\infty} f(x) e^{-x^2} dx$$

Functional derivative: generalization of derivative for functions.

$$F_v[\rho] = \int \rho(\vec{r}) v(\vec{r}) d\vec{r}$$
$$\frac{\delta F_v[\rho]}{\delta \rho} = v(\vec{r})$$

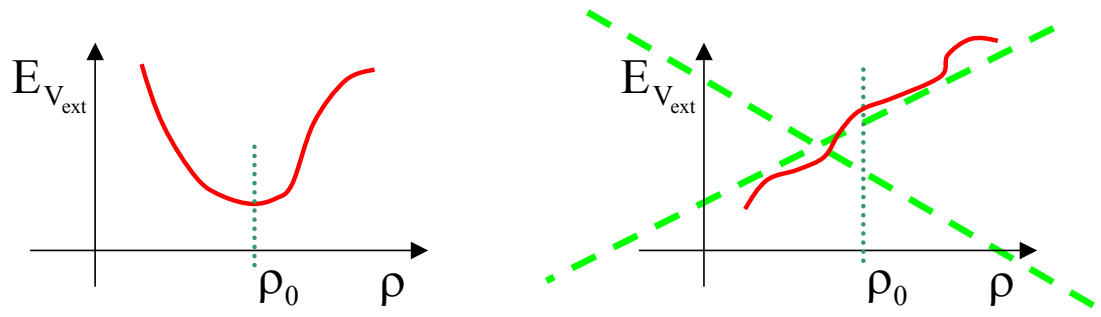
Second Theorem of Hohenberg and Kohn

“ The unique functional that returns the ground state total energy when applied to the ground state density, returns a higher energy for any other density. “



Second Theorem of Hohenberg and Kohn

“ The unique functional that returns the ground state total energy when applied to the ground state density, returns a higher energy for any other density. “



Note 1: this functional can be written as

$$E_{V_{ext}}[\rho] = F_{HK}[\rho] + \int \rho(\vec{r}) V_{ext}(\vec{r}) d\vec{r}$$

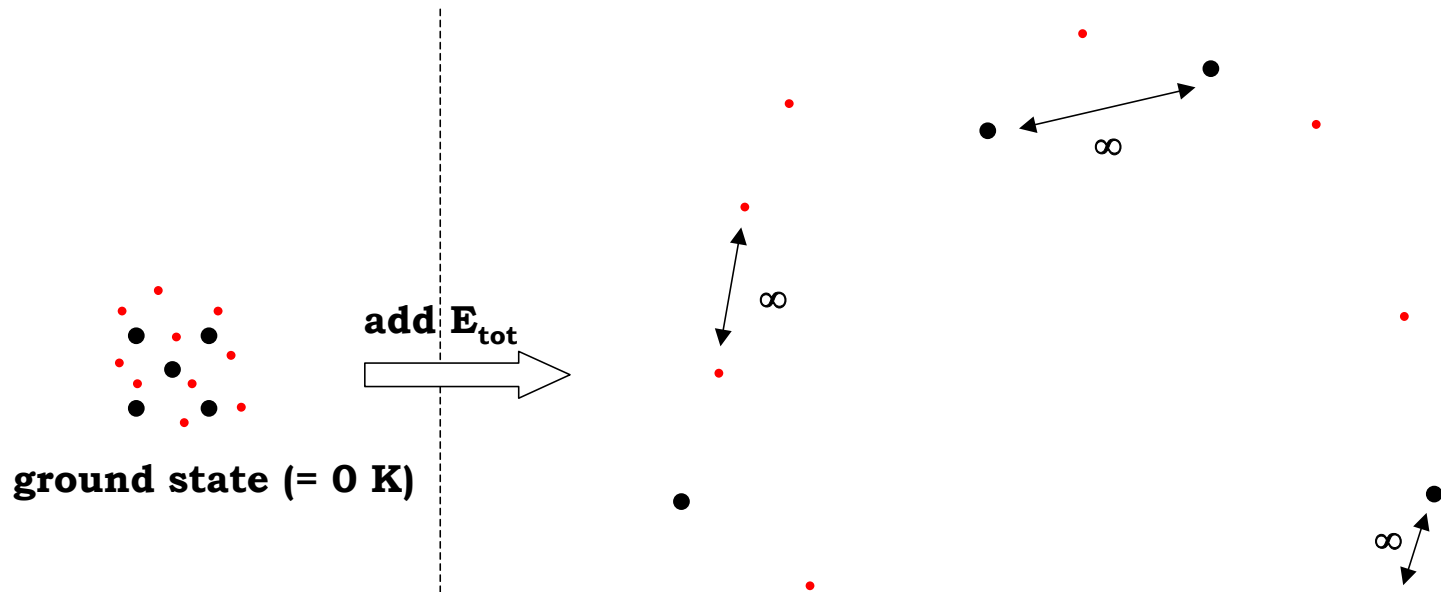
with F_{HK} the (unknown) Hohenberg-Kohn functional that returns the kinetic and electron-electron part of the total energy.

Note 2 : this suggests a procedure to find the ground state density by minimizing the total energy functional.

What is the “ground state total energy”

Ground state total energy =

The energy you have to add to the ground state to bring all nuclei and at infinite distance of each other.



E_{tot} can be experimentally determined from the specific heats, the latent heat of fusion and vaporization, and the energy needed to turn the remaining atoms into a plasma.

Theorem of Kohn and Sham

“ The exact ground state density of a N-electron system is

$$\rho(\vec{r}) = \sum_{i=1}^N \phi_i^*(\vec{r}) \phi_i(\vec{r})$$

where the single-particle wave functions ϕ_i are the N lowest-energy solutions of the Kohn-Sham equations :

$$\hat{H}_{\text{KS}} \phi_i = \varepsilon_i \phi_i$$

$$\begin{aligned} \text{with } \hat{H}_{\text{KS}} &= \hat{T}_0 + \hat{V}_{\text{H}} + \hat{V}_{\text{ext}} + \hat{V}_{\text{xc}} \\ &= -\frac{\hbar^2}{2m} \vec{\nabla}_i^2 + \frac{e^2}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}'|} d\vec{r}' + \hat{V}_{\text{ext}} + \hat{V}_{\text{xc}} \end{aligned}$$

$$\text{and } \hat{V}_{\text{xc}} = \frac{\delta V_{\text{xc}}[\rho]}{\delta \rho} .”$$

The Kohn-Sham equations are an exact transformation of the original problem (coupled differential equations) in single-particle equations (uncoupled)

Hartree-Fock

$$H = -\sum_i \frac{\nabla^2}{m_i} - \sum_{i,j} \frac{(eZ_i) e}{|\vec{R}_i - \vec{r}_j|} + \frac{1}{2} \sum_{i \neq j} \frac{e^2}{|\vec{r}_i - \vec{r}_j|}$$

- **electrons with kinetic energy**
 - **subject to Coulomb potential of nuclei**
 - **Coulomb-interacting with other electrons**
-

DFT

$$H = -\sum_i \frac{\nabla^2}{m_i} - \sum_{i,j} \frac{(eZ_i) e}{|\vec{R}_i - \vec{r}_j|} + V_H + V_{xc}$$

- **quasi-particles with kinetic energy**
- **subject to Coulomb potential of nuclei**
- **interacting with the average, static distribution of all other quasi-particles**
- **subject to the effect of a mysterious 'exchange-correlation' potential**

Much easier maths, but you have to know V_{xc}

Approximation: choose XC functional

One unique expression for $V_{xc}[\rho]$ must exist, but it is unknown.

Local Density Approximation (LDA) **postulates** that:

$$V_{xc}[\rho] = V_{xc}^{LDA}[\rho] = \int \rho(\vec{r}) \varepsilon_{xc}(\rho(\vec{r})) d\vec{r}$$

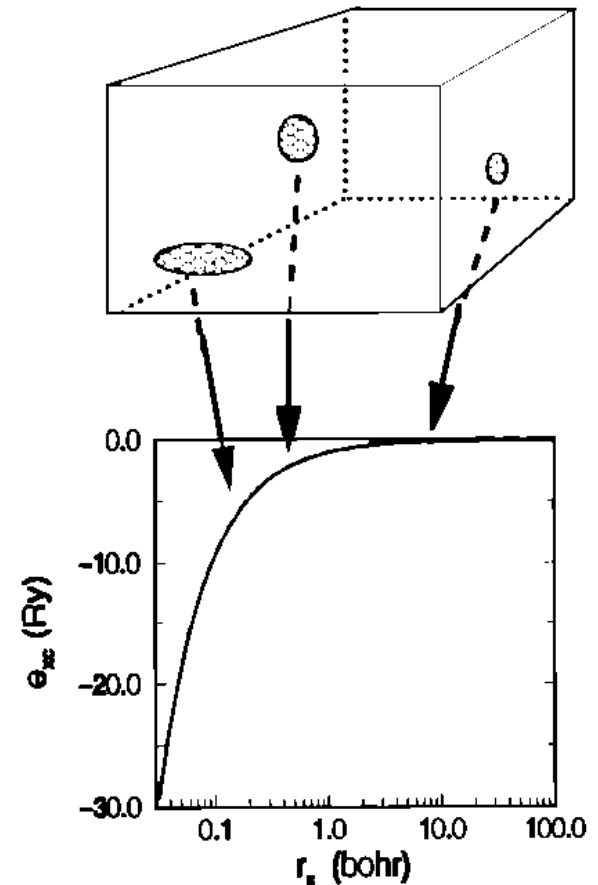
Local Spin Density Approximation (LSDA) **postulates** that:

$$V_{xc}[\rho] = V_{xc}^{LSDA}[\rho_{\uparrow}, \rho_{\downarrow}] = \int \rho(\vec{r}) \varepsilon_{xc}(\rho_{\uparrow}(\vec{r}), \rho_{\downarrow}(\vec{r})) d\vec{r}$$

Extension: Generalized Gradient Approximation GGA:

$$V_{xc}[\rho] = V_{xc}^{GGA}[\rho, \nabla\rho] = \int \rho(\vec{r}) \varepsilon_{xc}(\rho_{\uparrow}(\vec{r}), \rho_{\downarrow}(\vec{r}), \nabla\rho_{\uparrow}(\vec{r}), \nabla\rho_{\downarrow}(\vec{r})) d\vec{r}$$

Meta-GGAs also include second and higher derivatives of the density. This effectively amounts to building more and more accurate Taylor type approximations to the true XC functional.



Hybrid exchange-correlation functional: a mixture of LDA, GGA and Hartree-Fock exchange (which is known exactly):

$$E_{\text{XC}}^{\text{HYB}} = E_{\text{XC}}^{\text{LDA}} + a_0(E_{\text{X}}^{\text{HF}} - E_{\text{X}}^{\text{LDA}}) + a_{\text{X}}(E_{\text{X}}^{\text{GGA}} - E_{\text{X}}^{\text{LDA}}) + a_{\text{C}}(E_{\text{C}}^{\text{GGA}} - E_{\text{C}}^{\text{LDA}})$$

The coefficients are optimized against experimental data. This disqualifies DFT from the “*ab initio* methods” category – it does require empirical parameters.

TABLE V. Classical reaction barriers (in kcal/mol) computed using BLYP, HCTH, B3LYP, B97 and B97-1 functionals, compared to *ab initio* values.

Reaction	BLYP	HCTH	B3LYP	B97	B97-1	<i>Ab initio</i>
(I)	2.76	6.72	4.01	7.88	8.77	9.61 ^a
(II)	13.64	15.30	15.92	13.98	13.11	17.5, ^b 16.60 ^c
(III)	8.14	10.42	9.78	8.90	8.04	11.81 ^d
(IV)	3.51	6.09	6.33	5.81	5.05	9.51 ^d
(V)	...	1.32	1.23	1.31	0.67	5.62, ^d 4.87 ^c
(VI)	...	0.94	2.58	1.90	1.30	6.62, ^e 5.11 ^c

Do not just use B3LYP for everything – check the literature for what works best.

TABLE III. Optimized geometries (in Å and degrees) for selected training set molecules using BLYP, HCTH, B3LYP and B97-1 functionals, compared to reference values.

Molecule	Reference	BLYP	HCTH	B3LYP	B97-1
H ₂	0.741	0.743	0.742	0.740	0.739
LiH	1.596	1.598	1.616	1.589	1.596
BeH	1.343	1.346	1.360	1.341	1.349
CH	1.120	1.133	1.133	1.123	1.127
CH ₂ (³ B)	1.078	1.083	1.081	1.077	1.080
	136.0	135.7	135.2	135.3	135.1
CH ₂ (¹ A)	1.110	1.119	1.119	1.110	1.113
	101.9	101.9	100.6	101.8	101.4
CH ₃	1.080	1.084	1.081	1.078	1.080
CH ₄	1.086	1.094	1.091	1.088	1.090
NH	1.036	1.051	1.045	1.041	1.042
NH ₂	1.024	1.037	1.030	1.027	1.028
	103.3	102.5	102.2	103.1	102.9
NH ₃	1.012	1.020	1.014	1.013	1.014
	106.7	106.4	105.8	106.9	106.5
Δ <i>r</i>		0.019	0.013	0.008	0.008
Δφ		0.4	0.7	0.2	0.3

TABLE VI. Mean absolute errors (mae) for electron affinities in eV. G2-1 is the original G2 set (25 systems) (Refs. 17,18, and 20). Max ae denotes the maximum absolute error.

	Mae (G2-1)	Max ae(G2-1)
SVWN ^a	0.74	1.2
LSD (SVWN5)	0.30	0.7
PBE	0.11	0.3
BLYP ^a	0.11	0.4
B3LYP ^a	0.11	0.5
PBE1PBE	0.13	0.3

TABLE IX. Mean absolute error (mae) in Å for the bond lengths of the diatomics in Table VIII. Max ae denotes the maximum absolute error.

	Mae	Max ae
LSD (SVWN5)	0.016	0.095
PBE	0.014	0.054
BLYP ^a	0.019	0.045
VSXC ^a	0.013	0.075
B3LYP ^a	0.009	0.039
PBE1PBE	0.009	0.055

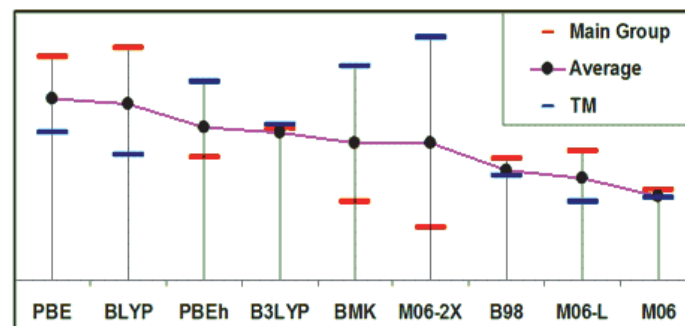
TABLE II. Mean absolute errors (mae) for atomization energies in kcal/mol, obtained from various DFT schemes. G2 denotes the extended G2 set (148 molecules) (Ref. 19) and G2-1 the original G2 set (55 molecules) (Refs. 17 and 18). The maximum absolute errors (max ae) are also listed.

	Mae (G2)	Mae (G2-1)	Max ae (G2)	Max ae (G2-1)
SVWN ^a	121.2	39.6	229	94
LSD(SVWN5)	83.7	36.4	216	84
PBE	17.1	8.6	52	26
BLYP ^a	7.1	4.7	28	15
VSXC ^b	2.7	2.5	12	8
B3LYP ^a	3.1	2.4	20	10
PBE1PBE	4.8	3.5	24	10

BLYP and B3LYP are the relatively recent success stories of DFT – Becke’s paper on the subject has collected over 10,000 citations.

TABLE IV. Mean absolute errors (mae) for the ionization potentials (in eV) in the G2-1 set Refs. (17,18 and 20) (the 38 atoms and molecules of Table III). Max ae denotes the maximum absolute error. In some cases self-consistent solutions could not be obtained (see Table III).

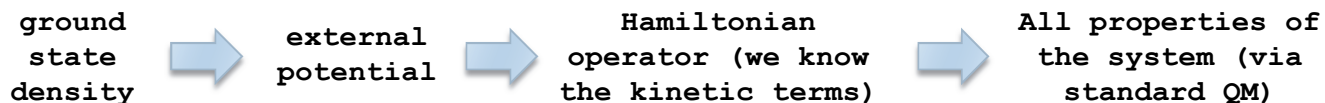
	Mae (G2-1)	Max ae(G2-1)
SVWN	0.69	1.2
LSD(SVWN5)	0.22	0.6
PBE	0.16	0.5
BLYP ^a	0.20	0.6
VSXC ^b	0.13	0.4
B3LYP ^a	0.17	0.8
PBE1PBE	0.16	0.7



Of the very recent exchange-correlation functional families, a very promising one is M06.

Summary of DFT

The basic lemma of DFT: *the ground state density of a bound system of interacting electrons in some external potential determines this potential uniquely.*



It should therefore be possible to reformulate quantum chemistry in terms of electron density alone (just three dimensions!) – a major simplification.

Hohenberg-Kohn variational principle: *there exists a well-defined (though not explicitly known) functional of electron density, which attains a global minimum at the ground state density.*

$$\min_{\rho} E[\rho] = \left\langle \Psi_0 \left| \hat{T} + \hat{V} + \hat{U} \right| \Psi_0 \right\rangle$$

Labels for the first equation:

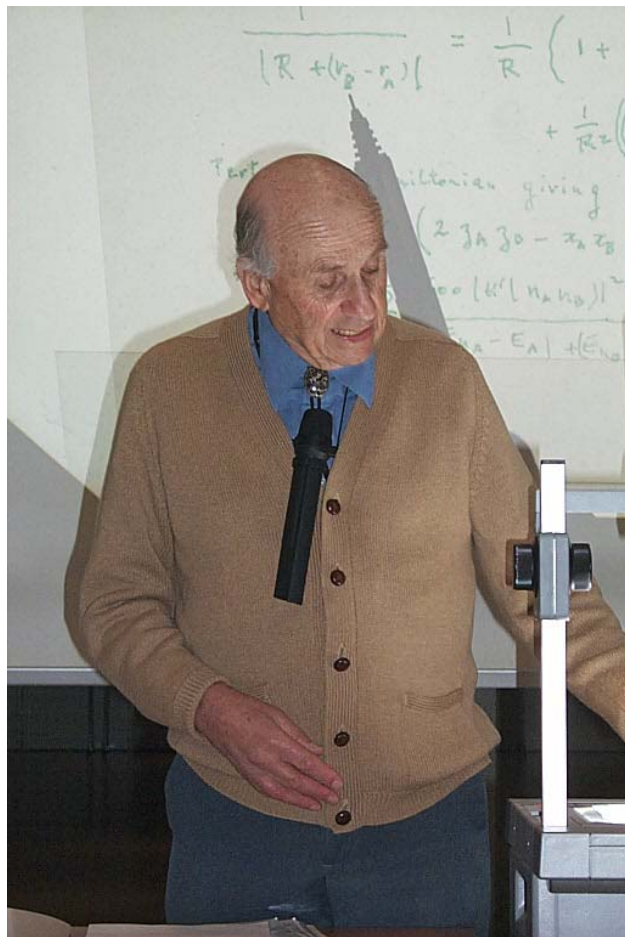
- ground state wavefunction (points to Ψ_0)
- kinetic terms (points to $\hat{T}$)
- nuclear attraction (points to $\hat{V}$)
- electron repulsion (points to $\hat{U}$)

$$E[\rho] = T[\rho] + U[\rho] + \underbrace{\int V(\vec{r})\rho(\vec{r})d^3\vec{r}}_{\text{nuclear attraction energy functional (known)}}$$

Labels for the second equation:

- energy functional (points to $E[\rho]$)
- kinetic energy functional (points to $T[\rho]$)
- inter-electron repulsion energy functional (points to $U[\rho]$)
- nuclear attraction energy functional (known) (points to the integral term)

Walter Kohn, “father of DFT”



Pictures from “Applied DFT 2001”
Vienna

Kinds of DFT functionals

Local Density Approximation (LDA):

$$V_{xc}^{LDA}[\rho] = \int \rho(\vec{r}) \varepsilon_{xc}(\rho(\vec{r})) d\vec{r} = -\frac{3}{4} \sqrt{\frac{3}{\pi}} \int \rho^{4/3}(\vec{r}) d\vec{r}$$

Generalised Gradient Approximation (GGA):

Becke (B88)

Lee, Yang, Parr (LYP)

$$\begin{aligned} \varepsilon_x^{\text{B88}} &= \varepsilon_x^{\text{LDA}} + \Delta\varepsilon_x^{\text{B88}} \\ \Delta\varepsilon_x^{\text{B88}} &= -\beta\rho^{1/3} \frac{x^2}{1 + 6\beta x \sinh^{-1} x} \\ x &= \frac{|\nabla\rho|}{\rho^{4/3}} \end{aligned}$$

$$\begin{aligned} \varepsilon_c^{\text{LYP}} &= -4a \frac{\rho_\alpha \rho_\beta}{\rho^2(1 + d\rho^{-1/3})} \\ &\quad - ab\omega \left\{ \frac{\rho_\alpha \rho_\beta}{18} \left[144(2^{2/3})C_F \left(\rho_\alpha^{8/3} + \rho_\beta^{8/3} \right) + (47 - 7\delta)|\nabla\rho|^2 \right. \right. \\ &\quad \left. \left. - (45 - \delta)(|\nabla\rho_\alpha|^2 + |\nabla\rho_\beta|^2) + 2\rho^{-1}(11 - \delta)(\rho_\alpha|\nabla\rho_\alpha|^2 + \rho_\beta|\nabla\rho_\beta|^2) \right] \right. \\ &\quad \left. + \frac{2}{3}\rho^2(|\nabla\rho_\alpha|^2 + |\nabla\rho_\beta|^2 - |\nabla\rho|^2) - \left(\rho_\alpha^2|\nabla\rho_\beta|^2 + \rho_\beta^2|\nabla\rho_\alpha|^2 \right) \right\} \\ \omega &= \frac{e^{-c\rho^{-1/3}}}{\rho^{14/3}(1 + d\rho^{-1/3})} \quad ; \quad \delta = c\rho^{-1/3} + \frac{d\rho^{-1/3}}{(1 + d\rho^{-1/3})} \end{aligned}$$

Meta-GGA: *TPSS*, *VSXC*, *HCTH*, etc. (include density, gradient of density and second derivative of density in the ε_{xc}

Kinds of DFT functionals

Hybrid Functionals: include some exchange from Hartree-Fock method (e.g. B3LYP)

$$E_{xc}^{B3LYP} = (1 - a)E_x^{LSDA} + aE_x^{HF} + b\Delta E_x^{B88} + (1 - c)E_c^{LSDA} + cE_c^{LYP}$$

Coefficients a , b , and c are found from fitting some set of experimental data.

Double Hybrid Functionals: include some HF exchange as well as some MP2-like energy (e.g. B2PLYP):

$$E_{xc}^{DHDFT} = (1 - a)E_x^{DFT} + aE_x^{HF} + (1 - b)E_c^{DFT} + bE_c^{MP2}$$

Coefficients a , b are found from fitting some set of experimental data.

The computation of E_c^{MP2} scales as N^5 , making the calculation quite slow for large systems.

Systematic improvability

- No problems for wave function based approaches:
just increase the CI expansion
- Much less clear for DFT
- Gradient corrected functional perform better than LSDA
- Hybrid functional are of comparable quality but the status is less clear (B3LYP is a “magic” functional)
- Across a wide variety of systems DFT tends to be remarkably robust
- Routes to further improve DFT: better defining the hole function; better correction of self-interaction errors

Worst-case scenarios

- Failure to describe London dispersion forces, originating from electron correlation at “long range”

Example: noble-gas dimers of He, Ne, etc., are predicted by DFT to be instable

- In hydrogen bonded systems, heavy-atom – heavy-atom is typically too short; errors for dissociation a relatively large in DFT
- For intermolecular complexes bound by charge transfer DFT overestimates the binding energy

Example: binding of ethylene to molecular fluorine is overestimated by 20 kcal/mol

- General problem with DFT: “overdelocalization”.

the self-interaction problem

in HF : Hartree energy =
$$\sum_{i,j} \int \frac{\rho_i(\vec{r})\rho_j(\vec{r}')}{|\vec{r}-\vec{r}'|} d\vec{r} d\vec{r}'$$

Exchange energy =
$$-\sum_{i,j} \int \frac{\psi_i^*(\vec{r})\psi_i(\vec{r}')\psi_j^*(\vec{r})\psi_j(\vec{r}')}{|\vec{r}-\vec{r}'|} d\vec{r} d\vec{r}'$$

cancellation of unphysical self-interaction :
$$\sum_{i,j} \int \frac{\rho_i(\vec{r})\rho_i(\vec{r}')}{|\vec{r}-\vec{r}'|} d\vec{r} d\vec{r}'$$

in DFT (Kohn-Sham) : Hartree energy =
$$\sum_{i,j} \int \frac{\rho_i(\vec{r})\rho_j(\vec{r}')}{|\vec{r}-\vec{r}'|} d\vec{r} d\vec{r}'$$

exactly cancelled by a term contained in the
(unknown) exact XC-functional

Practical versions of DFT suffer from a spurious, unphysical self-interaction contribution, leading to the need of 'self-interaction correction methods' (SIC). Especially important when electrons are densely packed and hence correlation being strong.

Kinds of DFT functionals

Range-Separated (long-range corrected) Functionals aim at addressing some systematic errors of DFT:

- **self-interaction errors** (for a given electron the Exchange cancel term does not cancel the Coulomb term exactly, leading to some error. In HF theory, Exchange matches exactly the Coulomb).
- excitation energies for charge-transfer states are too low (by TD-DFT)
- electron over-delocalisation problem

by partitioning the Coulomb operator term in 2 parts: *short* range and *long*-range:

$$\frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} = \frac{1}{r_{12}} = \frac{1 - \text{erf}(\omega r_{12})}{r_{12}} + \frac{\text{erf}(\omega r_{12})}{r_{12}}$$
$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

and apply the DFT exchange-correlation for short-range while using the exact HF exchange for the long-range.

Examples: LC-BLYP, CAM-B3LYP, etc.

Kinds of DFT functionals

Dispersion-corrected Functionals aim at accounting for various dispersion forces, like Van der Waals. These weak interactions are not accounted by standard DFT or HF methods, but proved increasingly important for large systems (it is cumulative).

Grimme proposed to include dispersion corrections as empirical terms, similar to Van Der Waals terms in Force Field methods.

Grimme:

$$\Delta E_{\text{disp}} = - \sum_{n=6(8,10)} s_n \sum_{\text{AB}}^{\text{atoms}} \frac{C_n^{\text{AB}}}{R_{\text{AB}}^n} f_{\text{damp}}(R_{\text{AB}})$$

Becke-Johnson (BJ):

$$C_6^{\text{AB}} = \frac{\alpha_A \alpha_B \langle \mu^2 \rangle_A \langle \mu^2 \rangle_B}{\alpha_A \langle \mu^2 \rangle_B + \alpha_B \langle \mu^2 \rangle_A}$$

Implementation depends tabulating (or evaluating) the parameters C_n^{AB} for any pair of atoms and the functional-dependent parameter s_n .

Dispersion may be included in the calculation by selecting the appropriate model. Grimme methods are denoted by D / D2 / D3 / D4 (e.g. ωB97D)

An alternative is to make ΔE_{disp} dependent on the actual density and evaluate it numerically (van-Voorhis-Vydrov). This is slower though and performance is similar to D3.

Overview & Performance of DFT

Table 6.1 Perdew classification of exchange–correlation functionals. *Italic* indicates that these include range-separated XC potentials while **bold** indicates inclusion of dispersion.

Level	Name	Variables	Examples	
			Few (≤ 12) parameters	Many (> 12) parameters
1	Local density	ρ	LDA, LSDA, X_α	
2	GGA	$\rho, \nabla \rho$	PW91, PBE, BLYP, OLYP, HCTH, KT3	SOGGA11, N12
3	Meta-GGA	$\rho, \nabla \rho, \nabla^2 \rho$ or τ	BR, B95, PKZB, TPSS, τ -HCTH, B97M-V	VSXC, M11L, MN12L
4	Hybrid-GGA	$\rho, \nabla \rho, \nabla^2 \rho$ or τ <i>HF exchange</i>	BHLYP, B3LYP, B3PW91, O3LYP, PBE0, TPSSh, τ -HCTH-hyb, HSE, <i>CAM-B3LYP</i> , ωB97X-V	B97-3, SOGGA11X, <i>M11</i> , <i>N12SX</i> , <i>MN12SX</i> , ω B97X, ωB97XD
5	Double hybrid	$\rho, \nabla \rho, \nabla^2 \rho$ or τ <i>HF exchange</i> <i>Virtual orbitals</i>	B2PLYP, PBE0-DH	

Table 6.2 Comparison of the performance of DFT methods in terms of Mean Absolute Deviations from reference values for different benchmarks (values in kJ/mol).

Functional	Ladder level ^a	MGAE109 ^b	SRMBE13	MRBE10	IP21	EA13	BH76
HF	–	129	144	304	92	113	48
MP2	–	9 (10)	22 (21)	161 (141)	27 (21)	13 (8)	20 (19)
CCSD	–	18 (8)	36 (26)	111 (93)	24 (12)	22 (11)	12 (10)
CCSD(T)	–	12 (1)	23 (14)	73 (43)	21 (8)	16 (2)	5 (2)
LSDA	1	70	89	127	41	24	63
BLYP	2	6	28	27	27	11	34
OLYP	2	4	39	27	12	15	23
PBE	2	13	29	60	26	9	37
HCTH407	2	5	27	37	28	15	25
SOGGA11	2	7	52	41	26	22	23
N12	2	5	38	30	14	18	29
τ -HCTH	3	4	40	44	19	9	27
TPSS	3	4	24	28	17	10	35
τ -HCTHhyb	4	3	28	36	17	8	20
TPSSh	4	4	19	68	14	12	27
B3LYP	4	4	25	91	24	10	18
PBE0	4	4	21	117	14	12	16
MN12L	4	3	48	31	15	11	7
CAM-B3LYP	4-RS	3	33	120	21	9	12
ω B97XD	4-RS	2	24	106	13	8	13
MN12-SX	4-RS	2	45	44	22	9	5
B2PLYP	5	4 (2)	18 (15)	74 (64)	10 (10)	10 (6)	8 (8)

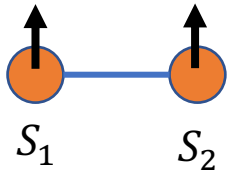
^aLadder level: 1 = local spin density approximation, 2 = generalized gradient approximation, 3 = meta generalized gradient approximation, 4 = hybrid functional including HF exchange, 4-RS = 4 with range-separated exchange, 5 = double hybrid method. All results refer to calculations using a TZP-type basis set. The values in parenthesis are from extrapolation to the complete basis set limit.

^bMAD per bond.

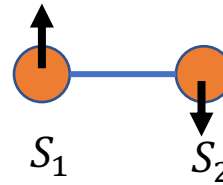
Some advices

- 1) Think about the wave function of the investigated system and assess whether DFT is applicable at all (i.e., do you expect near-degeneracy of energy levels?). Most closed-shell organic molecules are suitable for DFT. All open-shell mono-radicals -- as well (unrestricted DFT). Bi-radicals and poly-radicals – less suitable. Transition metal compounds – less suitable, unless they are clearly closed-shell.
- 2) Lanthanides are suitable only for 4f-in-core pseudopotentials.
- 3) Acquire as much experimental data as possible so you can compare calculations with experiment. Are the measurements done in solvent or in the gas phase?
- 4) Choose the basis sets. Start with a small basis, then project the results onto a larger basis (many codes do this automatically). Anions require extra diffuse functions and range-corrected functionals. Usually VTZP basis gives reasonable results, and larger bases do not bring extra accuracy.
- 5) For small systems, you may neglect dispersion. For large systems, dispersion is important.
- 6) If your system contains a heavy element (e.g., Pt, Pd, Zn, etc.), then relativistic corrections are important. Search for DKH or ZORA keywords.
- 7) Read about the best-performing DFT functional & basis sets for your particular problem and class of compounds. Finding literature is faster than performing the benchmark study by yourself.

Magnetic Couplings



Ferro-magnetic interaction



Anti-ferro magnetic interaction

$$S_1 = S_2 = \frac{1}{2}$$

True wave functions for:

Triplet $S=1$:

$$\Psi_{S=1, M=+1} = \alpha(1)\alpha(2)$$

$$\Psi_{S=1, M=0} = \frac{1}{\sqrt{2}} [\alpha(1)\beta(2) + \beta(1)\alpha(2)]$$

$$\Psi_{S=1, M=-1} = \beta(1)\beta(2)$$

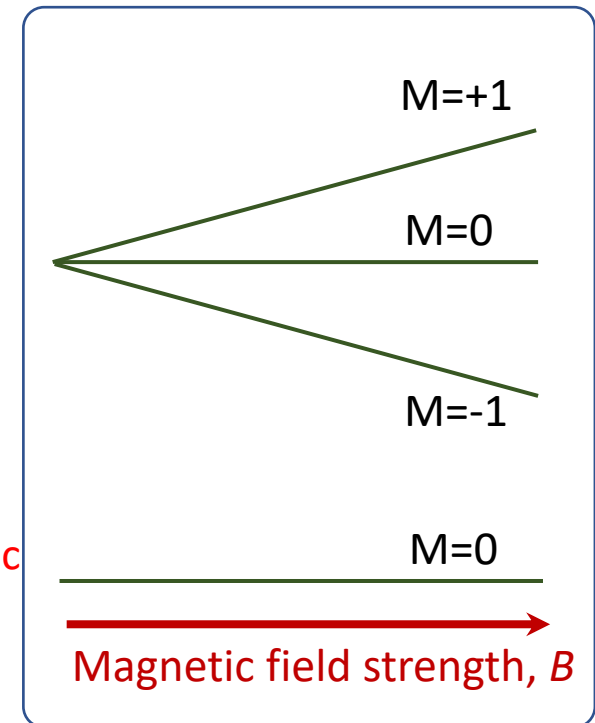


magnetic

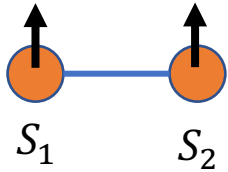
Singlet $S=0$:

$$\Psi_{S=0, M=0} = \frac{1}{\sqrt{2}} [\alpha(1)\beta(2) - \beta(1)\alpha(2)]$$

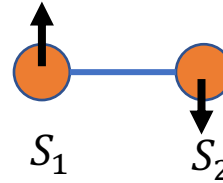
non-magnetic



Magnetic Couplings



Ferro-magnetic interaction
 $J > 0$



Anti-ferro magnetic interaction
 $J < 0$

$$S_1 = S_2 = \frac{1}{2}$$

Magnetic interaction is described phenomenologically by Heisenberg-Dirac-Van-Vleck Hamiltonian :

$$\hat{H}_{exch} = -2J\hat{S}_1\hat{S}_2$$

Where J is the parameter describing the strength of magnetic interaction.

It can be re-written using the total spin: $S_T: |S_1 + S_2|, |S_1 + S_2 - 1|, \dots |S_1 - S_2|$

$$S_T^2 = (S_1 + S_2)^2 = S_1^2 + S_2^2 + 2\hat{S}_1\hat{S}_2; \quad 2\hat{S}_1\hat{S}_2 = S_T^2 - S_1^2 - S_2^2$$

$$\hat{H}_{exch} = -J[S_T^2 - S_1^2 - S_2^2]$$

Now, this Hamiltonian can be expanded in the basis of eigenstates of $S = 1$ and $S = 0$ (as given on previous slide).

Magnetic Couplings

$$\hat{H}_{exch} = -J[S_T^2 - S_1^2 - S_2^2]$$

Now, this Hamiltonian can be expanded in the basis of eigenstates of $S=1$ and $S=0$ (as given on previous slide).

$$\begin{aligned} \text{triplet: } E(S_T = 1) &= -J[S_T(S_T + 1) - S_1(S_1 + 1) - S_2(S_2 + 1)] = -J\left[2 - \frac{1}{2} \times \frac{3}{2} - \frac{1}{2} \times \frac{3}{2}\right] = -\frac{1}{2}J \\ \text{singlet: } E(S_T = 0) &= -J[S_T(S_T + 1) - S_1(S_1 + 1) - S_2(S_2 + 1)] = -J\left[0 - \frac{1}{2} \times \frac{3}{2} - \frac{1}{2} \times \frac{3}{2}\right] = +\frac{3}{2}J \end{aligned}$$

Verification:

$$-\frac{1}{2}J \times (2 \times 1 + 1) + \frac{3}{2}J \times (2 \times 0 + 1) = -\frac{3}{2}J + \frac{3}{2}J = 0$$

Now, setting $S = 0$ as ground state (i.e. at 0), we get the energy of the triplet as:

$$E(S_T = 0) = 0$$

$$E(S_T = 1) = \Delta E_{ST} = -\frac{1}{2}J - \left(+\frac{3}{2}J\right) = -2J$$

If $J > 0 \rightarrow E(S_T = 1) < E(S_T = 0) \rightarrow$ ferromagnetic coupling.

The energy gap between the singlet and triplet is directly linked to the parameter of magnetic interaction.

If $E(S_T = 1)$ and $E(S_T = 0)$ are estimated computational methods, then J could be readily derived. No issues for CI or multireference methods here, except that the correlated calculations are tedious.

Magnetic Couplings from DFT

Can we evaluate the parameter of magnetic interaction using DFT?

Triplet $S=1$:

$$\Psi_{S=1,M=+1} = \alpha(1)\alpha(2)$$

→ one Slater Determinant, suitable for DFT

$$\Psi_{S=1,M=0} = \frac{1}{\sqrt{2}}[\alpha(1)\beta(2) + \beta(1)\alpha(2)]$$

$$\Psi_{S=1,M=-1} = \beta(1)\beta(2)$$

Singlet $S=0$:

$$\Psi_{S=0,M=0} = \frac{1}{\sqrt{2}}[\alpha(1)\beta(2) - \beta(1)\alpha(2)]$$

→ two Slater Determinant, NOT suitable for DFT

As result, the direct DFT might not be suitable for computing the singlet state.

Broken symmetry (BS) approach was proposed to overcome this issue. In BS-DFT, the singlet wave function is not computed, but instead, the energy of one single determinant $\alpha(1)\beta(2)$ is evaluated. Parameter J is derived from the energy difference of two configurations:

$$\begin{aligned} E_{HS} - E_{BS} &= [\langle 1|h|1 \rangle + \langle 2|h|2 \rangle + \langle 12|12 \rangle - \langle 12|21 \rangle]_{HS} - [\langle 1|h|1 \rangle + \langle 2|h|2 \rangle + \langle 12|12 \rangle]_{BS} \\ &= -\langle 12|21 \rangle = -\left\langle \alpha(1)\beta(2) \left| \frac{1}{r_{12}} \right| \alpha(2)\beta(1) \right\rangle = -2J \end{aligned}$$

$$J = \frac{E_{HS} - E_{BS}}{2} = \frac{\langle 12|21 \rangle}{2} = \frac{\left\langle \alpha(1)\beta(2) \left| \frac{1}{r_{12}} \right| \alpha(2)\beta(1) \right\rangle}{2}$$

Magnetic Couplings from DFT

For more general cases when $S_{1,2} \geq \frac{1}{2}$ the above formula is not quite correct, as the magnetic coupling depends on the values of the interacting spin and therefore must consider the larger value of the total spin.

Yamaguchi et al. [Chem. Phys. Lett. 432, (2006), 343–347.] proposed the following expression:

$$J = \frac{E_{BS} - E_{HS}}{\langle S^2 \rangle_{BS} - \langle S^2 \rangle_{HS}} = \frac{E_{HS} - E_{BS}}{\langle S^2 \rangle_{HS} - \langle S^2 \rangle_{BS}}$$

where $\langle S^2 \rangle_{BS}$ and $\langle S^2 \rangle_{HS}$ are the expectation values of the S^2 operator on the respective wave functions, while E_{BS} and E_{HS} are the total energies of the broken-symmetry and high-spin configurations.

Alternative formulas for magnetic coupling J is given by Noodleman. et. al.

ORCA software allows for an easy-to-use implementation of the BS-DFT for magnetic coupling. Several formulas are implemented.

A research study using the ORCA for this purpose: Phys.Chem.Chem.Phys., 2014, 16, 14568.

GAUSSIAN software also allows computing the magnetic coupling. See a tutorial here:

<https://gaussian.com/afc/>

Propagator Methods with DFT

If the molecule is subjected to a linear fluctuating electric field, $\mathbf{E} = \mathbf{r} \cos \omega t$, then the frequency dependent polarizability is well approximated by:

$$\langle \alpha \rangle_{\omega} = \sum_{i=0}^{Nstates} \frac{|\langle \psi_0 | \mathbf{r} | \psi_i \rangle|^2}{\omega - (E_i - E_0)}$$

Propagator methods (based on **Green's function** or **Equation of Motion** (EOM)) aim at finding the frequency dependent polarizability without explicit account of the ψ_i .

Random Phase Approximation (RPA) method accounts for the effect of single and double electron excitations onto the CI matrix elements, without explicitly building the CI wave functions. RPA can be applied to HF wave function or to CASSCF. Dynamic electron correlation accounted by RPA is of medium quality.

Time dependent DFT is an equivalent of the RPA for a DFT wave function and aims at optimizing the excited states by considering that the excitation of one electron from some occupied orbital to the virtual orbital is like an external perturbation to the original system.

Of course, there is no actual “time-dependency” in the excited state calculations using TD-DFT. The term originates from the fact that TD-DFT equations started from solving the time-dependent Schrodinger equation, with subsequent making the so-called adiabatic approximation, which is equivalent to “losing time memory”, i.e. no time-dependency.

TD-DFT

TD-DFT works well for the following conditions:

- a) the ground state wave function is well described by a single Slater determinant
- b) low-values of spin contamination
- c) only low-lying electronic excited states are required
- d) suitable basis set and density integration grids are used
- e) suitable dispersion correction methods are used

Not all possible combinations of available solvent effects, relativistic corrections, dispersion corrections are available for TD-DFT in all software packages. Some bugs are also expected; therefore it is recommended to analyze the results critically alongside with suitable testing.

High excited states which are comparable to molecular ionization potentials are poorly described.

Practical session: ORCA software

Useful links for learning the ORCA software:

- 1) orca_manual_5_0_2.pdf (download from Canvas: Files -> Examples_for_various_calculations -> ORCA)
- 2) ORCA Input Library: <https://sites.google.com/site/orcainputlibrary/home>
- 3) ORCA Forum: <https://orcaforum.kofo.mpg.de/index.php>
- 4) ORCA Tutorial site: <https://www.faccts.de/docs#orca>

General Input:

<https://www.youtube.com/watch?v=-lndIVOvcW0>
<https://www.youtube.com/c/IaNiusha>
<https://www.youtube.com/watch?v=TqSmmnqXWF8>

UVVIS Spectroscopy:

https://www.orcasoftware.de/tutorials_orca/spec/UVVis.html

Magnetic Coupling:

<https://www.youtube.com/watch?v=G32950pVtO8>
https://www.youtube.com/watch?v=uE_UYjuEuRQ

Transition State:

<https://www.youtube.com/watch?v=FycOHebcOhs>
<https://www.youtube.com/watch?v=v8rqV1rFCNU>

Practical session: ORCA software on HPC

```
! UKS B3LYP def2-SVP def2/J TightSCF TightOpt Freq LARGEPRINT
```

```
%pal nprocs 12 end  
%maxcore 1000
```

```
%scf  
  MaxIter 1000  
end
```

```
%output  
  XYZFile true  
end
```

```
* xyz 0 1  
C   0.00000   1.40272   0.00000  
H   0.00000   2.49029   0.00000  
C  -1.21479   0.70136   0.00000  
H  -2.15666   1.24515   0.00000  
C  -1.21479  -0.70136   0.00000  
H  -2.15666  -1.24515   0.00000  
C   0.00000  -1.40272   0.00000  
H   0.00000  -2.49029   0.00000  
C   1.21479  -0.70136   0.00000  
H   2.15666  -1.24515   0.00000  
C   1.21479   0.70136   0.00000  
H   2.15666   1.24515   0.00000  
*
```

Practical session

Main Task : Find the magnetic exchange for the $[\text{Cu}_2\text{Cl}_6]^{2-}$ compound.

Subtask: Inspect the molecule with Avogadro or an equivalent program

Subtask: Prepare the input and compute the $S=1$ state. Inspect the results.

> find on which atoms are the unpaired electrons residing

> find the spin contamination index

Subtask: Prepare the input for “SpinFlip”, i.e. telling the program to reverse the spin on one of the Cu atoms

Subtask: Launch the BS-DFT calculation. Inspect the results. Locate on which atoms are the unpaired electrons residing. Was the spin actually flipped on the desired atom? What is the spin contamination for the BS wave function?

Subtask: Compare the magnetic couplings of the attached $[\text{Cu}_2\text{Cl}_6]^{2-}$ systems.

Extra: If you wish, the magnetic exchange can be calculated as function of some structural deformations. e.g. Cu-Cl-Cu angle, angle by which the outer two Cl make with the Cu_2Cl_2 plane, etc.